

Change the title of Clause 162 (added by IEEE Std 802.3ck-2022) as follows:

162. Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4, and 800GBASE-CR8

162.1 Overview

Change the first three paragraphs of 162.1 as follows:

This clause specifies the 100GBASE-CR1 ~~PMD~~, the 200GBASE-CR2 ~~PMD~~, the 400GBASE-CR4, and 800GBASE-CR8 ~~PMDs~~, and the baseband medium. The specifications for the ~~three~~ four ~~PMDs~~ are similar, except for the number of lanes and associated parameters and the MDI.

There are four associated annexes. Annex 162A provides information on parameters ~~with test points that may that might~~ not be testable in an implemented system, since the test points they are associated with are typically inaccessible. Annex 162B specifies test fixtures. Annex 162C specifies MDIs. Annex 162D describes host and cable assembly types.

When forming a complete Physical Layer, a PMD shall be connected as illustrated in Figure 162–1, to the appropriate sublayers (as specified in [Table 162–1](#), [Table 162–2](#), ~~and Table 162–3~~, and Table 162–3a), to the medium through the appropriate MDI, and to the management functions that are optionally accessible through the management interface defined in Clause 45, or equivalent.

Insert new Table 162–3a after Table 162–3 as follows:

Table 162–3a—Physical Layer clauses associated with the 800GBASE-CR8 PMD

Associated clause	800GBASE-CR8
170—RS	Required
170—800GMII ^a	Optional
171—800GMII Extender	Optional
172—PCS for 800GBASE-R	Required
173—PMA for 800GBASE-R	Required
120F—800GAUI-8 C2C	Optional
73—Auto-Negotiation	Required

^a The 800GMII is an optional interface. However, if the 800GMII is not implemented, a conforming implementation behaves functionally as though the RS and 800GMII were present.

Insert the following paragraph after the fifth paragraph of 162.1:

For the 800GBASE-CR8 PHY, in order to support the required frame loss ratio (see [1.4.344](#)) of less than 6.2×10^{-11} for 64-octet frames with minimum interpacket gap, the PMD and the adjacent PMA are expected to detect bits from a compliant input signal at a BER lower than 2.4×10^{-4} assuming errors are sufficiently uncorrelated. This BER allocation enables a frame loss ratio lower than 3.4×10^{-12} after processing by the

PCS (Clause 172) if there are negligible errors due to other electrical interfaces (800GAUI-n). If the PMD and PMA create errors that are not sufficiently uncorrelated, the BER is required to be lower as appropriate to maintain a frame loss ratio lower than 3.4×10^{-12} .

Change last paragraph of 162.1 as follows:

Figure 162–1 shows the relationship of the ~~100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4~~ PMD sublayers and MDIs to the ISO/IEC Open System Interconnection (OSI) reference model.

Replace Figure 162–1 with the following figure:

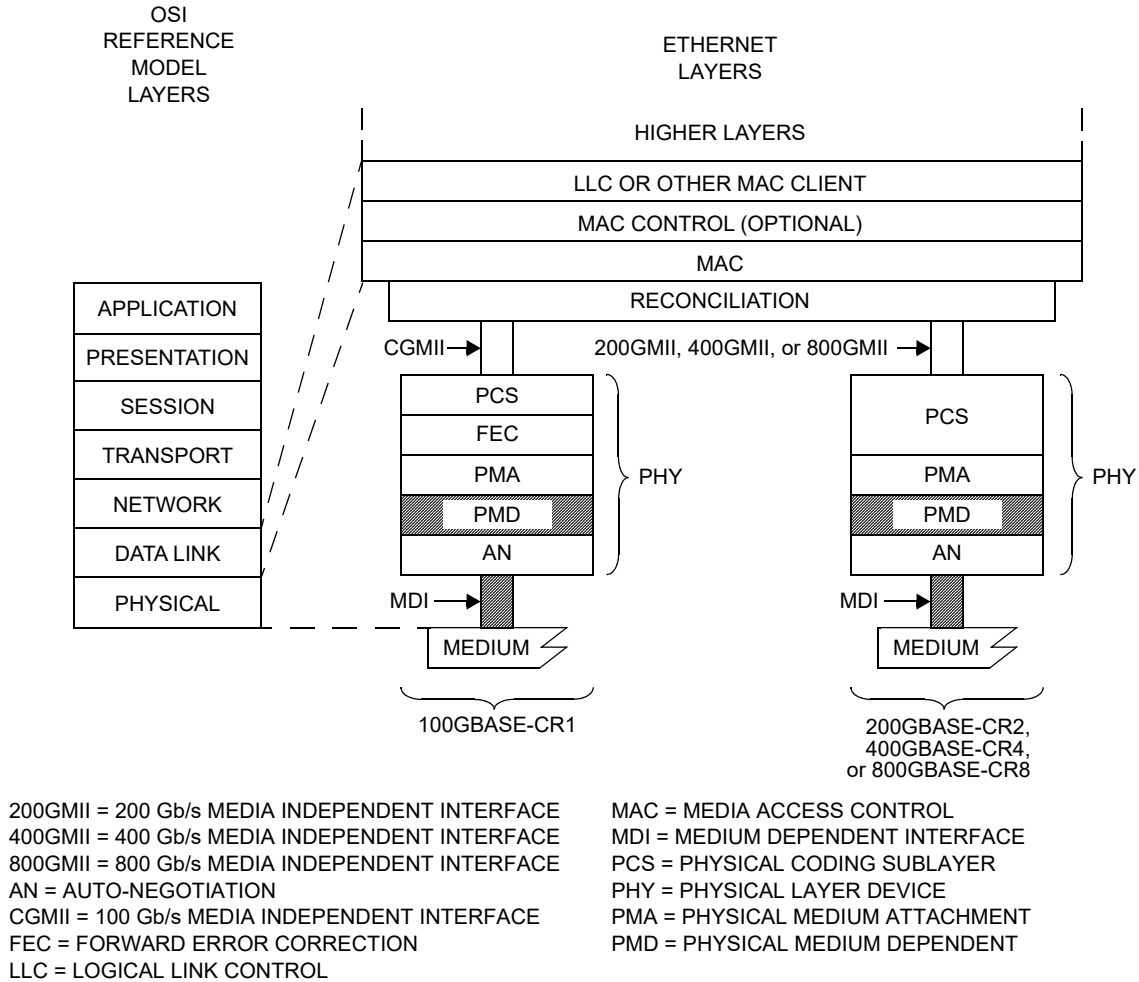


Figure 162–1—100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, and 800GBASE-CR8 relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model

162.2 Conventions

Change 162.2 as follows:

Clause 162 describes ~~three-four~~ PMDs; 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4, and 800GBASE-CR8~~, which have one, two, ~~and four, and eight~~ lanes, respectively. For efficient description, the parameter n is used to describe the number of lanes in a specific PMD. Accordingly, $n = 1$ for 100GBASE-CR1, $n = 2$ for 200GBASE-CR2, ~~and $n = 4$ for 400GBASE-CR4, and $n = 8$ for 800GBASE-CR8.~~

The parameter i is used as an index or a suffix to identify a specific lane, and takes the values 0 to $n - 1$.

Within this clause, the unqualified term “PMD” refers to any of 100GBASE-CR1 PMD, 200GBASE-CR2 PMD, ~~or 400GBASE-CR4 PMD, or 800GBASE-CR8 PMD.~~

Within this clause, the term FEC refers to either [Clause 91](#) RS-FEC or ~~the~~ [Clause 161](#) RS-FEC-Int for 100GBASE-CR1, ~~or the RS-FEC within the~~ [Clause 119](#) PCS for 200GBASE-CR2 and 400GBASE-CR4, ~~or the RS-FEC within the~~ [Clause 172](#) PCS for 800GBASE-CR8.

162.3 PMD service interfaces

Change the first paragraph of 162.3 as follows:

This subclause specifies the services provided by the 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4, and 800GBASE-CR8~~ PMDs. The service interfaces for these PMDs are described in an abstract manner and do not imply any particular implementation. The PMD service interface supports the exchange of encoded data between the PMA entity that resides just above the PMD, and the PMD entity. The PMD translates the encoded data to and from signals suitable for the specified medium.

Insert the following paragraph after the fourth paragraph of 162.3:

The 800GBASE-CR8 PMD service interface is an instance of the inter-sublayer service interface defined in 169.3, with eight parallel symbol streams ($n = 8$).

Insert the following paragraph after the eighth (now ninth) paragraph of 162.3:

The 800GBASE-CR8 PMD has eight parallel symbol streams, hence $i = 0$ to 7.

162.4 PCS requirements for Auto-Negotiation (AN) service interface

Change the text in 162.4 as follows:

The PCSs associated with the ~~100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4~~ PMDs are required to support the AN service interface primitive AN_LINK.indication defined in [73.9](#). (See [82.6](#), ~~and~~ [119.6](#), ~~and~~ [172.6](#).)

100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4, and 800GBASE-CR8~~ PHYs may be extended using CAUI-n/100GAUI-n C2C, 200GAUI-n C2C, ~~and 400GAUI-n C2C, and 800GAUI-n C2C~~, respectively, as physical instantiations of the inter-sublayer service interface between devices. If such extension is used, then the AN_LINK(link_status).indication is relayed from the device with the PCS sublayer to the device with the AN sublayer by means at the discretion of the implementer. As examples, the

implementer may employ use of pervasive management or employ a dedicated electrical signal to relay the state of link_status as indicated by the PCS sublayer on one device to the AN sublayer on the other device.

162.5 Delay constraints

Change Table 162–4 as follows:

Table 162–4—Delay constraints

PMD	Maximum (bit times) ^a	Maximum (pause_quanta) ^b	Maximum (ns)
100GBASE-CR1	4 096	8	40.96
200GBASE-CR2	8 192	16	40.96
400GBASE-CR4	16 384	32	40.96
<u>800GBASE-CR8</u>	<u>32 768</u>	<u>64</u>	<u>40.96</u>

^a One bit time is equal to 10 ps for 100GBASE-CR1, 5 ps for 200GBASE-CR2, ~~and 2.5 ps for 400GBASE-CR4, and~~ 1.25 ps for 800GBASE-CR8. (See 1.4.215 for the definition of bit time.)

^b One pause_quantum is equal to 5.12 ns for 100GBASE-CR1, 2.56 ns for 200GBASE-CR2, ~~and 1.28 ns for~~ 400GBASE-CR4, and 0.64 ns for 800GBASE-CR8. (See 31B.2 for the definition of pause_quanta.)

162.6 Skew constraints

Insert new subclause 162.6.3 after 162.6.2 as follows:

162.6.3 Skew Constraints for 800GBASE-CR8

Skew and Skew Variation are defined in 169.5 and specified at the points SP1 to SP6 shown in Figure 169–5.

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, then the Skew at SP2 is limited to 25 ns and the Skew Variation at SP2 is limited to 400 ps.

The Skew at SP3 (the transmitter MDI) shall be less than 36 ns and the Skew Variation at SP3 shall be less than 600 ps.

The Skew at SP4 (the receiver MDI) shall be less than 116 ns and the Skew Variation at SP4 shall be less than 3.4 ns.

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, then the Skew at SP5 shall be less than 127 ns and the Skew Variation at SP5 shall be less than 3.6 ns.

For more information on Skew and Skew Variation see 169.5.

162.7 PMD MDIO function mapping

Change Table 162–5, Table 162–6, and Table 162–7, including footnotes, as follows:

Table 162–5—MDIO/PMD control variable mapping

MDIO control variable	PMA/PMD register name	Register/bit number	PMD control variable
Reset	PMA/PMD control 1	1.0.15	PMD_reset
Global PMD transmit disable	PMD transmit disable	1.9.0	Global_PMD_transmit_disable
<u>PMD transmit disable 7 to PMD transmit disable 4^a</u>	<u>PMD transmit disable</u>	<u>1.9.8:5</u>	<u>PMD_transmit_disable_7,</u> <u>PMD_transmit_disable_6,</u> <u>PMD_transmit_disable_5,</u> <u>PMD_transmit_disable_4</u>
PMD transmit disable 3 to PMD transmit disable 2 ^b	PMD transmit disable	1.9.4:3	PMD_transmit_disable_3, PMD_transmit_disable_2
PMD transmit disable 1 ^c	PMD transmit disable	1.9.2	PMD_transmit_disable_1
PMD transmit disable 0	PMD transmit disable	1.9.1	PMD_transmit_disable_0

^a Available only in 800GBASE-CR8.

^b Available only in 400GBASE-CR4 and 800GBASE-CR8.

^c Available only in 200GBASE-CR2, and 400GBASE-CR4, and 800GBASE-CR8.

Table 162–6—MDIO/PMD status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMD status variable
Fault	PMA/PMD status 1	1.1.7	PMD_fault
Transmit fault	PMA/PMD status 2	1.8.11	PMD_transmit_fault
Receive fault	PMA/PMD status 2	1.8.10	PMD_receive_fault
Global PMD receive signal detect	PMD receive signal detect	1.10.0	Global_PMD_signal_detect
<u>PMD receive signal detect 7 to PMD receive signal detect 4^a</u>	<u>PMD receive signal detect</u>	<u>1.10.8:5</u>	<u>PMD_signal_detect_7,</u> <u>PMD_signal_detect_6,</u> <u>PMD_signal_detect_5,</u> <u>PMD_signal_detect_4</u>
PMD receive signal detect 3 to PMD receive signal detect 2 ^b	PMD receive signal detect	1.10.4:3	PMD_signal_detect_3, PMD_signal_detect_2
PMD receive signal detect 1 ^c	PMD receive signal detect	1.10.2	PMD_signal_detect_1
PMD receive signal detect 0	PMD receive signal detect	1.10.1	PMD_signal_detect_0

^a Available only in 800GBASE-CR8.

^b Available only in 400GBASE-CR4 and 800GBASE-CR8.

^c Available only in 200GBASE-CR2, and 400GBASE-CR4, and 800GBASE-CR8.

Table 162–7—MDIO/PMD control function mapping

MDIO variable	PMA/PMD register name	Register/bit number	PMD variable
Restart training	BASE-R PMD control	1.150.0	mr_restart_training
Training enable	BASE-R PMD control	1.150.1	mr_training_enable
Receiver status	BASE-R PMD status	1.151.0 ^a	local_trained
Frame lock	BASE-R PMD status	1.151.1 ^a	local_tf_lock
Start-up protocol status	BASE-R PMD status	1.151.2 ^a	training
Training failure	BASE-R PMD status	1.151.3 ^a	training_failure
Polynomial identifier	PMD training pattern, lanes 0 to 3 7	1.1450.12:11 ^b	identifier_i
Seed	PMD training pattern, lanes 0 to 3 7	1.1450.15:14 1.1450.10:0 ^b	seed_i
Initial condition request	BASE-R PAM4 PMD training LP control, lanes 0 to 3 7	1.1120.13:12 ^b	ic_req
Coefficient select	BASE-R PAM4 PMD training LP control, lanes 0 to 3 7	1.1120.4:2 ^b	coef_sel
Coefficient request	BASE-R PAM4 PMD training LP control, lanes 0 to 3 7	1.1120.1:0 ^b	coef_req
Receiver ready	BASE-R PAM4 PMD training LP status, lanes 0 to 3 7	1.1220.15 ^b	remote_rx_ready
Modulation and precoding status	BASE-R PAM4 PMD training LP status, lanes 0 to 3 7	1.1220.11:10 ^b	remote_tp_mode
Receiver frame lock	BASE-R PAM4 PMD training LP status, lanes 0 to 3 7	1.1220.9 ^b	remote_tf_lock
Modulation and precoding request	BASE-R PAM4 PMD training LD control, lanes 0 to 3 7	1.1320.11:10 ^b	local_tp_mode
Local receiver ready	BASE-R PAM4 PMD training LD status, lanes 0 to 3 7	1.1420.15 ^b	local_rx_ready
Initial condition status	BASE-R PAM4 PMD training LD status, lanes 0 to 3 7	1.1420.8 ^b	ic_sts
Coefficient select echo	BASE-R PAM4 PMD training LD status, lanes 0 to 3 7	1.1420.5:3 ^b	k
Coefficient status	BASE-R PAM4 PMD training LD status, lanes 0 to 3 7	1.1420.2:0 ^b	coef_sts

^a Bit reference is provided for lane 0. Status bits for lanes 1 to 3 are located in the same register. Status bits for lanes 4 to 7 are located in BASE-R PMD status 2 register (Register 1.156).

^b Address reference is provided for lane 0. Registers for lanes 1 to ~~3~~ 7 are located at an offset from the lane 0 register.

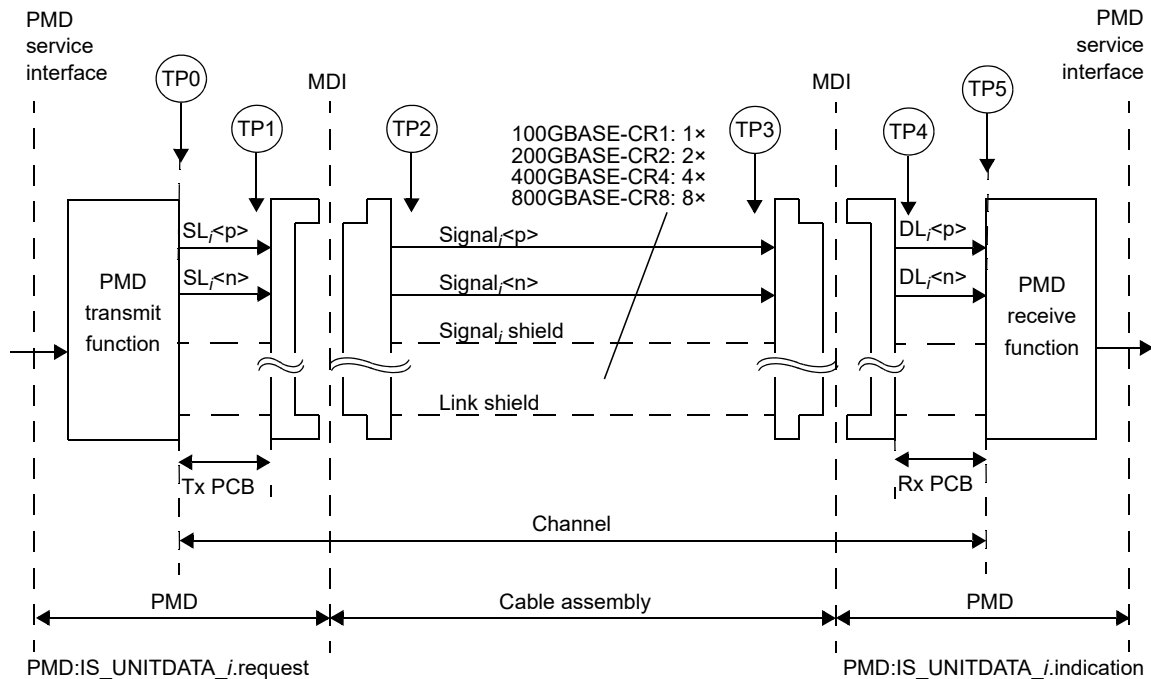
162.8 PMD functional specifications

162.8.1 Link block diagram

Change the first paragraph of 162.8.1 as follows:

One direction of a 100GBASE-CR1, 200GBASE-CR2, ~~or 400GBASE-CR4,~~ or 800GBASE-CR8 link is shown in Figure 162–2. The PMDs on both ends of the link have connected ground references. Each differential pair in the cable is separately shielded as illustrated in Figure 162–2. The signal shields are connected to ground contacts in the MDI plug connectors on both ends of the cable assembly.

Replace Figure 162–2 with the following figure:



NOTE—One direction is illustrated.

Figure 162–2—100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, or 800GBASE-CR8 link

162.8.11 PMD control function

Insert a new item h) at the end of the lettered list in 162.8.11, as follows:

- h) The default polynomials and seeds for the training pattern (see 136.8.11.1.3) on each lane are defined in 162.8.11.1.

Insert new subclause 162.8.11.1 at the end of 162.8.11 as follows:

162.8.11.1 Training pattern polynomials and seeds

The PRBS generator for each lane shall implement four generator polynomials. The polynomial used in each lane *i* is selected by the variable identifier *i*.

At the start of the training pattern in each lane i , the state of the PRBS generator shall be set to the value of the variable seed_i . A value of all zeros is not valid.

Table 162–10a specifies the default identifier, the corresponding polynomial, and the recommended default value of seed_i for each lane i , as well as the first 13 symbols of the training pattern for each modulation and precoding mode created using the default polynomial and seed.

The training pattern is generated from the PRBS generator output as specified in 136.8.11.1.3.

Table 162–10a—Training pattern default polynomials and seeds

i	Default identifier	Polynomial, $G(x)$	Default seed $_i^a$	Initial output, PAM2	Initial output, PAM4	Initial output, PAM4 with precoding
0	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	0000010101011	00303303330000	1031320220111	1301200200101
1	1	$1 + x^2 + x^3 + x^7 + x^{13}$	0011101000001	3030303030333	3030213021333	3122012201212
2	2	$1 + x^2 + x^4 + x^8 + x^{13}$	1001000101100	0303333033030	1212332133031	1102120121301
3	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0100010000010	3330300030330	2231210121221	2032013201110
4	0 ^b	$1 + x + x^2 + x^{12} + x^{13}$	1111110100110	3030000303303	3030001313212	3122223012011
5	1	$1 + x^2 + x^3 + x^7 + x^{13}$	1100011101110	0003030003033	0113130013133	0103213103212
6	2	$1 + x^2 + x^4 + x^8 + x^{13}$	0000001101000	3300303000300	2300212111300	2131102323000
7	3	$1 + x^2 + x^5 + x^9 + x^{13}$	0011000100111	3333000333030	2232000322031	2033131202210

^a The leftmost bit in the sequence corresponds to the initial value of S0 and the rightmost bit corresponds to the initial value of S12 in Figure 136–5.

^b This training pattern is equivalent to the PRBS13Q test pattern defined in 120.5.11.2.1.

NOTE—The set of polynomials, and the default seed bits for identifiers 0 through 3, are identical to those specified in Table 136–8.

162.9 PMD electrical characteristics

162.9.2 Signal paths

Change the second paragraph of 162.9.2 as follows:

For 100GBASE-CR1, there is one differential lane in each direction for a total of two pairs, or four connections. For 200GBASE-CR2, there are two differential lanes in each direction for a total of four pairs, or eight connections. For 400GBASE-CR4, there are four differential lanes in each direction for a total of eight pairs, or ~~sixteen~~ 16 connections. For 800GBASE-CR8, there are eight differential lanes in each direction for a total of 16 pairs, or 32 connections.

162.9.4 Transmitter characteristics

Change footnote a in Table 162–11 as follows (unchanged rows and footnotes not shown):

Table 162–11—Summary of transmitter specifications at TP2

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range)	—	53.125 ± 50 ppm ^a	GBd
...			

^a For a 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 PMD in the same package as the PCS sublayer or for any 800GBASE-CR8 PMD. In other cases, the signaling rate is derived from the input to the PMD transmit function provided by the adjacent PMA sublayer.

162.9.5 Receiver characteristics

Change Table 162–15 as follows (most unchanged rows not shown):

Table 162–15—Summary of receiver specifications at TP3

Parameter	Subclause reference	Value	Units
Signaling rate, each lane (range) 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4 800GBASE-CR8	162.9.5.1	53.125 ± 100 ppm 53.125 ± 50 ppm	GBd GBd
...			
Differential-mode to common-mode return loss, RL_{cd} (min)	162.9.5.6	Equation (162–18)	dB
NOTE—For 100GBASE-CR1, 200GBASE-CR2, or 400GBASE-CR4, although the PMD transmitter is specified with a signaling rate range of ±50 ppm when in the same package as the PCS, the signaling rate range may be ±100 ppm when derived from an intermediate interface (e.g., 100GAUI-4).			

162.9.5.1 Receiver signaling rate

Change 162.9.5.1 as follows:

A PHY shall comply with the receiver requirements of 162.9.5.2, 162.9.5.3, and 162.9.5.4 for any signaling rate in the range specified in Table 162–15 53.125 GBd ± 100 ppm.

NOTE—Although the PMD transmitter is specified with a signaling rate range of ±50 ppm when in the same package as the PCS sublayer, the signaling rate range may be ±100 ppm, when derived from an intermediate interface (e.g., 100GAUI-4).

162.10 Channel characteristics

Change the second paragraph of 162.10 as follows:

Channel definitions apply for links between two PHYs of the same type: 100GBASE-CR1, 200GBASE-CR2, ~~or 400GBASE-CR4,~~ or 800GBASE-CR8.

162.11 Cable assembly characteristics

Change the text in 162.11 as follows:

Cable assemblies defined in this subclause contain insulated conductors terminated in a connector at each end for use as link segments between MDIs. Cable assemblies are primarily intended as point-to-point links between 100GBASE-CR1, 200GBASE-CR2, ~~or 400GBASE-CR4,~~ or 800GBASE-CR8 PHYs using controlled impedance cables.

~~Three~~ Four cable assembly types are specified:

- a) 100GBASE-CR1: Cable assembly that supports single-lane links between two 100GBASE-CR1 PHYs with achievable cable length of at least 2 m.
- b) 200GBASE-CR2: Cable assembly that supports two-lane links between two 200GBASE-CR2 PHYs with achievable cable length of at least 2 m.
- c) 400GBASE-CR4: Cable assembly that supports four-lane links between two 400GBASE-CR4 PHYs with achievable cable length of at least 2 m.
- d) 800GBASE-CR8: Cable assembly that supports eight-lane links between two 800GBASE-CR8 PHYs with achievable cable length of at least 2 m.

NOTE—It may be possible to construct compliant cable assemblies longer than indicated. Length of a cable assembly does not imply compliance to specifications.

100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4,~~ and 800GBASE-CR8 cable assembly types are described in Annex 162D. The MDIs are defined in Annex 162C.

For 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4,~~ and 800GBASE-CR8, the lanes are AC-coupled. The AC-coupling shall be within the cable assembly. It is recommended that it is within the plug connectors. It should be noted that there may be various methods for AC-coupling in actual implementations. The low-frequency 3 dB cutoff of the AC-coupling shall be less than 50 kHz. It is recommended that the value of the coupling capacitors be 100 nF. The capacitor limits the inrush charge and baseline wander.

All cable assembly measurements are to be made between TP1 and TP4 with cable assembly test fixtures as specified in Annex 162B. These cable assembly specifications are based upon twinaxial cable characteristics, but other cable types are acceptable if the specifications of this subclause are met.

Table 162–18 provides a summary of the cable assembly characteristics for 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4,~~ and 800GBASE-CR8, and references to the subclauses addressing each parameter.

The specifications for the ~~three~~ four cable assembly types are identical except the number of lanes.

162.12 MDI specifications

Change the text in 162.12 as follows:

The MDI couples the PMD (specified in 162.8 and 162.9) to the cable assembly (specified in 162.11).

100GBASE-CR1 has six specified MDI connectors: SFP112, SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

200GBASE-CR2 has five specified MDI connectors: SFP-DD112, DSFP, QSFP112, QSFP-DD800, and OSFP.

400GBASE-CR4 has three specified MDI connectors: QSFP112, QSFP-DD800, and OSFP.

800GBASE-CR8 has two specified MDI connectors: QSFP-DD800 and OSFP.

Annex 162C specifies the MDIs for 100GBASE-CR1, 200GBASE-CR2, 400GBASE-CR4, and 800GBASE-CR8.

Change the heading of 162.14 as follows:

**162.14 Protocol implementation conformance statement (PICS) proforma for
Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium,
type 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4~~, and 800GBASE-CR8⁸**

162.14.1 Introduction

Change the first paragraph of 162.14.1 as follows:

The supplier of a protocol implementation that is claimed to conform to Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4~~, and 800GBASE-CR8, shall complete the following protocol implementation conformance statement (PICS) proforma.

162.14.2 Identification

162.14.2.2 Protocol summary

Change the table in 162.14.2.2 as follows:

Identification of protocol standard	IEEE Std 802.3df-2024, Clause 162, Physical Medium Dependent (PMD) sublayer and baseband medium, type 100GBASE-CR1, 200GBASE-CR2, and 400GBASE-CR4 , <u>and 800GBASE-CR8</u>
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3df-2024.)	

Date of Statement	
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⁸Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

162.14.3 Major capabilities/options

Change the table in 162.14.3 as follows (some unchanged rows not shown):

Item ^a	Feature	Subclause	Value/Comment	Status	Support
...					
*CR4	400GBASE-CR4 PMD	162.1	Can operate as a 400GBASE-CR4 PMD	O.1	Yes [] No []
*CR8	<u>800GBASE-CR8 PMD</u>	<u>162.1</u>	<u>Can operate as an 800GBASE-CR8 PMD</u>	<u>O.1</u>	<u>Yes []</u> <u>No []</u>
PMA100	100GBASE-P PMA	162.1	Device implements <u>Clause 135</u> PMA for 100GBASE-P	CR1:M	Yes [] N/A []
...					
PMA400	400GBASE-R PMA	162.1	Device implements Clause 120 PMA for 400GBASE-R	CR4:M	Yes [] N/A []
<u>PMA800</u>	<u>800GBASE-R PMA</u>	162.1	<u>Device implements Clause 173 PMA for 800GBASE-R</u>	<u>CR8:M</u>	<u>Yes []</u> <u>N/A []</u>
*AUIPMD	100GAUI-n C2C, 200GAUI-n C2C, or 400GAUI-n C2C, <u>or 800GAUI-n C2C</u>	162.1	Service interface of PMA adjacent to PMD is physically instantiated	O.2	Yes [] No []
AUIFEC	CAUI-n C2C or 100GAUI-n C2C	162.1	Service interface of PMA between PCS and RS-FEC is physically instantiated	CR4:O.2	Yes [] No [] N/A []
*nGMII	CGMII, 200GMII, or 400GMII, <u>or 800GMII and RS</u>	162.1	Interface is implemented or functionally equivalent	O.2	Yes [] No []
...					
PCS400	400GBASE-R PCS	162.1	Device implements <u>Clause 119</u> 400GBASE-R PCS	CR4* nGMII:M	Yes [] N/A []
<u>PCS800</u>	<u>800GBASE-R PCS</u>	162.1	<u>Device implements Clause 172 800GBASE-R PCS</u>	<u>CR8*</u> <u>nGMII:M</u>	<u>Yes []</u> <u>N/A []</u>
PCSAN	PCS requirements for AN service interface	162.4	PCS supports service interface primitive AN_LINK.indication	nGMII:M	Yes [] N/A []
...					

^a A “*” preceding an “Item” identifier indicates there are other PICS that depend on whether or not this item is supported.

Change the title of 162.14.4 as follows:

**162.14.4 PICS proforma tables for Physical Medium Dependent (PMD) sublayer and
 baseband medium, type 100GBASE-CR1, 200GBASE-CR2, ~~and 400GBASE-CR4, and~~
800GBASE-CR8**

162.14.4.2 PMD control function

*Change the rows for items PC2 and PC3 in the table in 162.14.4.2 as follows (unchanged rows not
 shown):*

Item	Feature	Subclause	Value/Comment	Status	Support
...					
PC2	Training pattern	136.8.11.1.3 162.8.11.1	Each lane implements four generator polynomials defined in Table 136–8 Table 162–10a	M	Yes []
PC3	Training pattern	136.8.11.1.3 162.8.11.1	State set to the value of seed_ <i>i</i> at the start of the training pattern	M	Yes []
...					